

Distributed Optimization for Machine Learning

Lecture 17 - Robust distributed optimization: Attacks and defenses

ECE 5290/7290 & ORIE 5290

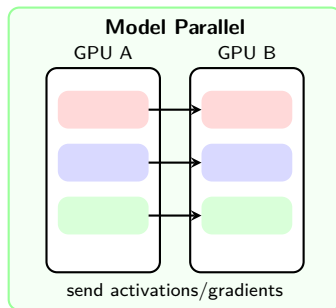
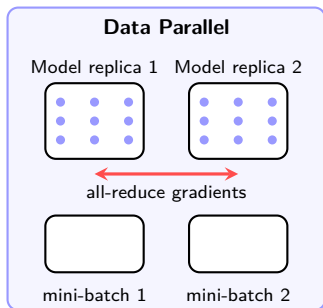
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November 3, 2025



Recall two paths of distributed learning

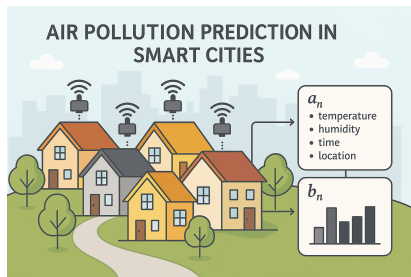


Data parallel: replicate the model; split the data; *consensus* at each step.

Model parallel: split the model across devices; exchange activations.



Example: air pollution indicator in smart homes



There is a community of multiple houses, where each house has a smart sensor that records environmental data $\xi_n = \{a_n, b_n\}$ over time, where:

$a_n = (\text{temperature, humidity, time, location, etc.})$

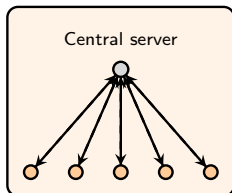
$b_n = \text{concentration of a particular pollutant.}$

Goal: Collaboratively train a model to predict b given a : $b = h(x; a)$.



Review of Parallel SGD for distributed learning

$$\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}), \quad \text{where } f_i(\mathbf{x}) = \mathbb{E}_{\xi_i \sim D_i}[F(\mathbf{x}; \xi_i)].$$



Local data on nodes

$$\begin{aligned} g_i^k &= \nabla F(\mathbf{x}^k; \xi_i^k) && \text{(Local compt.)} \\ \mathbf{x}^{k+1} &= \mathbf{x}^k - \frac{\eta}{n} \sum_{i=1}^n g_i^k && \text{(Global comm.)} \end{aligned}$$

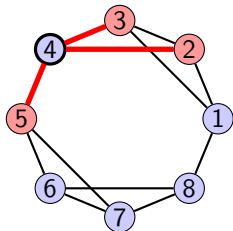
- All nodes synchronize (i.e., *globally average*) **every time**.

Limitations: Communication cost is $\mathcal{O}(n)$ which is high when n is large.



Last-lecture: Decentralized SGD for distributed learning

$$\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}), \quad \text{where } f_i(\mathbf{x}) = \mathbb{E}_{\xi_i \sim D_i} [F(\mathbf{x}; \xi_i)].$$



Every node has 3 neighbors.

$$\mathbf{x}_i^{k+\frac{1}{2}} = \mathbf{x}_i^k - \eta \nabla F(\mathbf{x}_i^k; \xi_i^k)$$

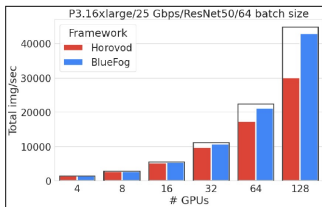
$$\mathbf{x}_i^{k+1} = \sum_{j \in \mathcal{N}_i} w_{ij} \mathbf{x}_j^{k+\frac{1}{2}}$$

- **Idea:** Instead of global averaging, partial averaging with neighbors

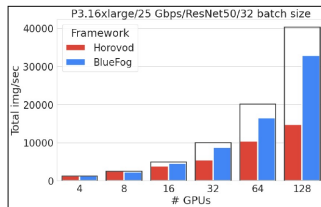
Benefit and limitation: Reduce comm. cost to $O(d_{\max})$ but need to design a double-stochastic mixing matrix.



The power of distributed learning



Small comm.-to-compt. ratio



Large comm.-to-compt. ratio

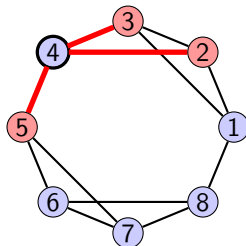
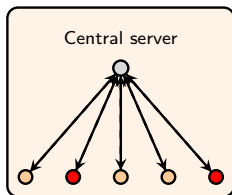
Example: data parallel training across multiple compute nodes.

- Often achieve linear speedup as the number of nodes increases
- Decentralization helps reduce communication overhead
- Preserve data privacy by keeping raw data local at each node



What if some nodes are unreliable?

The aggregation rule combines gradients or models into an update. The power of distributed learning depends on **trusting each node's update**.

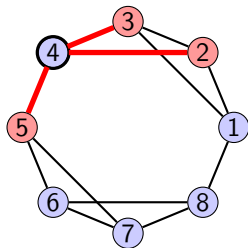
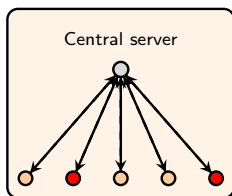


But what if some nodes are **unreliable** or **malicious**?

- Smart homes - faulty or spoofed devices send incorrect readings.
- Mobile keyboards - adversarial clients inject biased updates.
- Autonomous systems - compromised vehicles or sensors.



Scope of this lecture



But what if some nodes are **unreliable** or **malicious**?

Goal of this lecture:

- G1)** Understand how malicious behavior or attacks can corrupt learning.
- G2)** Explore defense mechanisms via robust aggregation.
- G3)** Investigate how non-i.i.d. data and topology impact robustness (★).



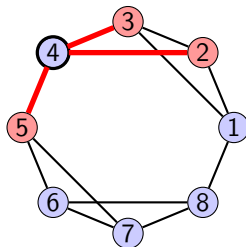
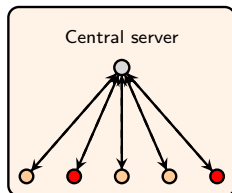
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Vulnerability of Distributed Learning

Byzantine-Robust Aggregation Strategies



The attack model - Byzantine attack



What an **unreliable** or **malicious** node can do?

Byzantine attacks are a type of malicious attack in distributed systems where one or more participants act as "traitors" or "Byzantine nodes," sending arbitrary information to prevent the system's consensus.



Historical origin: Where does “Byzantine” come from?

- The term originates from the **Byzantine Empire** (330 - 1453 CE), centered in modern-day Turkey, which was renowned for its *complex governance, intrigue, and layers of communication*.



The Byzantine Empire (centered around modern-day Istanbul).

- Introduced by (Lamport et al' 1982) in “*The Byzantine Generals Problem*” to describe agreement under faulty or malicious nodes.



The metaphor: Consensus under conflicting information

- Imagine generals coordinating an attack from different locations.
- Some may be **traitors**, sending **inconsistent or false messages**.
- The honest generals must still reach a **common, correct decision**.



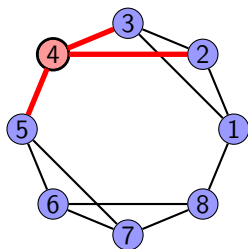
Honest vs. traitorous generals trying to reach agreement.



Interactive game: consensus under Byzantine nodes

Scenario: Not all nodes behave honestly.

- You are a **node** in a peer network with a number x_i (**cannot change**).
- Share your number with neighbors.
- Compute an aggregate via *Simple Average* with 2-3 neighbors.
- **One or two nodes are Byzantine** and may send fake numbers; they do not follow the update rule.



● Honest node ● Byzantine node

How much damage can a small number of Byzantine nodes cause?



Two mini-experiments (4 - 5 rounds each)

Experiment 1 (Average):

1. Everyone uses **mean** of received values.
2. After 5 rounds, scan the QR code and upload the values for each round.



mean aggregation

Experiment 2 (Median):

1. Everyone uses **median** of received values.
2. After 5 rounds, scan the QR code and upload the values for each round.



median aggregation

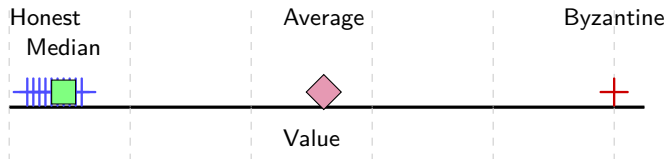
Observation task:



Which aggregation resists outliers better - Average or Median?

Takeaways from consensus games

- Did the **average** drift toward extreme values?
- Did the **median** stay close to the honest cluster?



Outlier inflates the average; the median stays near the majority.

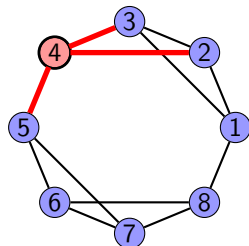
Key takeaway: Average is fragile; robust aggregators (e.g., median) resist Byzantine outliers.



From the consensus game to distributed learning

The difference between consensus and distributed learning?

- Each node now sends vector **gradients or model updates**, not scalars.
- The consensus values are dynamic and **changing iteratively**.
- Reach consensus on **the convergent model**, not intermediate results.



● Honest node ● Byzantine node

Key insight: The same fragility we saw in average consensus reappears in distributed learning with mean gradient aggregation.



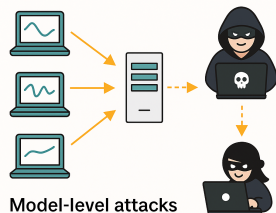
Two categories of Byzantine attacks

Model-level attacks (or model poisoning)

- Corrupt gradients or parameters.
 - Add random Gaussian noise.
 - Flip the sign of gradients.
 - Scale updates to dominate.
 - Craft gradients to mimic honest nodes.

Data-level attacks (or data poisoning)

- Poison local data before training.
 - Label-flipping
 - Add random Gaussian noise.
 - Backdoor insertion.



Example: Crafted gradient (model-level) attack

- **How it works:** The attacker (e.g., node m) generates $\tilde{g}_m^k$ (via local SGD on a targeted objective) and sends $\delta_m^k = \lambda \tilde{g}_m^k$ to server, so the server sees $\tilde{g}_m^k$ effectively scaled in the average.

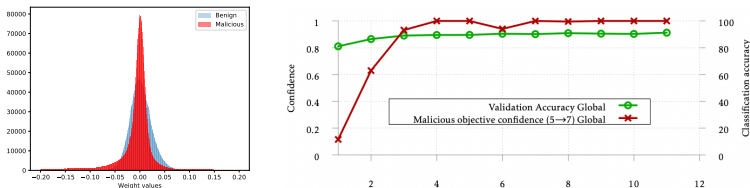


Figure: Model poisoning for CNN on Fashion MNIST.

- **Notes:** Effective attacks choose λ and $\tilde{g}_m^k$ so that $\tilde{g}_m^k$ statistics (norm/variance/sparsity) resemble honest g_i^k .



Example: Backdoor (data-level) attack

- **How it works:** Attacker injects a small trigger pattern into local data samples, and re-labels those samples to a chosen target class.
- Example: Images with a corner patch are labeled as “stop sign”.

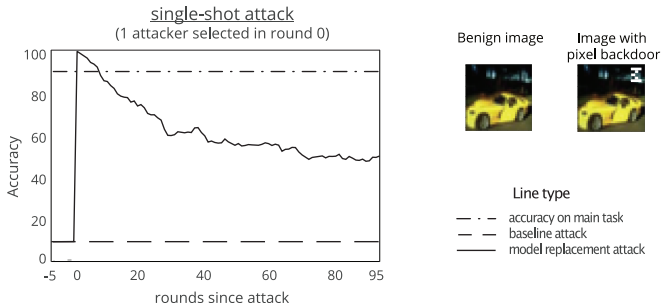


Figure: Backdoored model misclassifies images with a custom pattern as birds.



Bagdasaryan, A. Veit, Y. Hua, D. Estrin, V. Shmatikov. “How to Backdoor Federated Learning”, *AISTATS 2020*

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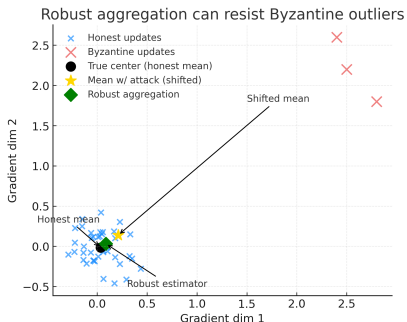


Can we still learn despite malicious nodes?

Goal: Design **robust aggregation rules** so that:

Aggregated update $\approx$ average of honest nodes

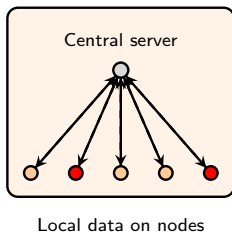
even if among n nodes, up to αn nodes are Byzantine.



Replace average aggregation with robust aggregation

Setting: n nodes, a set of Byzantine nodes $\mathcal{B}$ and of honest nodes $\mathcal{H}$

$\mathcal{G} = \{\mathbf{g}_1, \dots, \mathbf{g}_n\}$, with at most αn adversarial gradients.



Goal: Design a robust aggregator $\mathcal{A}(\mathcal{G})$ to estimate the true mean of **honest nodes** $\mathcal{H}$

$$\mathcal{A}(\mathcal{G}) \approx \mathbf{g}^* = \frac{1}{|\mathcal{H}|} \sum_{i \in \mathcal{H}} \mathbf{g}_i$$

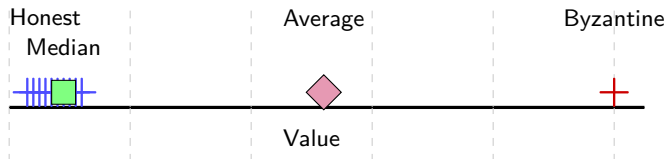
without knowing which ones are corrupted.

Server updates the global model by the robust aggregator $\mathcal{A}(\mathcal{G})$:

$$\mathbf{x}^{k+1} = \mathbf{x}^k - \eta \mathcal{A}(\mathcal{G}^k).$$



Why average aggregation fails?



Recall: Average consensus or gradient aggregation is fragile.

- A single outlier (malicious node) can distort the result.
- The average is highly sensitive to extreme values.

Idea: Use **robust statistics** that are less influenced by outliers.

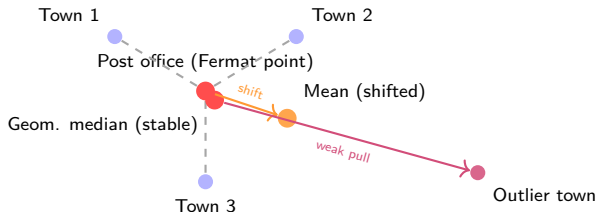


Fermat's problem: where to build the post office?



Pierre de Fermat,
17th-century

- Pierre de Fermat asked “Given three towns, where should a post office be built so that the total travel distance to all towns is minimized?”
- This point minimizes total distance - known as the **Fermat point**, or **geometric median**.

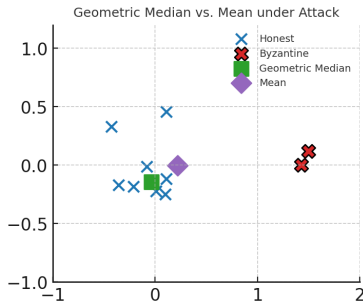


Geometric median - generalization of median

Definition of geometric median:

$$\mathcal{A}(\mathcal{G}) = \arg \min_{z \in \mathbb{R}^d} \sum_{i=1}^n \|g_i - z\|$$

it finds the point minimizing the total Euclidean distance to others.



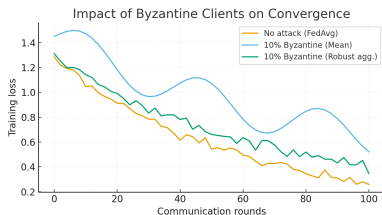
Properties:

- In $d = 1$, this reduces to the **median**, $\min_z \sum_i |g_i - z|$.
- **At the optimum** z^* , each g_i has a unit force: $\sum_{i=1}^n \frac{z^* - g_i}{\|z^* - g_i\|} = 0$.



Impact of Byzantine attacks on optimization dynamics

- **Setup (simulated):** n nodes, 10% Byzantine.
- **Observation:** Mean aggregation collapses (loss stalls or spikes).
- **Robust rules:** Geometric median track the baseline closely.



Clean (baseline), mean+attack, GeoMedian.



Upper bound of the robust aggregation error

Theorem 1 (Bounded aggregation error)

If at least $(1 - \alpha)n$ honest vectors in $\mathcal{G} = \{\mathbf{g}_1, \dots, \mathbf{g}_n\}$ satisfy

$$\frac{1}{(1 - \alpha)n} \sum_{i \in \mathcal{H}} \|\mathbf{g}_i - \mathbf{g}^*\| \leq \sigma, \quad \text{for some } \alpha < \frac{1}{2},$$

where $\mathbf{g}^*$ is the ideal direction among *all honest nodes*, then

$$\|\mathcal{A}(\mathcal{G}) - \mathbf{g}^*\| \leq C_\alpha \sigma, \quad \text{with } C_\alpha = 1 + \frac{1}{1 - 2\alpha}.$$

- C_α : robustness - accuracy tradeoff; remains small for moderate α .
- σ : input variation (e.g., data variance) of honest nodes
- **Guarantee:** The aggregated update is close to the mean if $\alpha < 0.5$.



Proof of Theorem 1*

- Using the definition $\mathcal{A}(\mathcal{G}) = \arg \min_{\mathbf{z}} \sum_{i=1}^n \|\mathbf{g}_i - \mathbf{z}\|$, we have

$$\sum_{i=1}^n \|\mathbf{g}_i - \mathcal{A}(\mathcal{G})\| \leq \sum_{i=1}^n \|\mathbf{g}_i - \mathbf{g}^*\| \quad (\diamond)$$

- Now investigate the error: $\|\mathcal{A}(\mathcal{G}) - \mathbf{g}^*\|$, by the triangle inequality

$$\|\mathbf{g}_i - \mathcal{A}(\mathcal{G})\| \geq \|\mathcal{A}(\mathcal{G}) - \mathbf{g}^*\| - \|\mathbf{g}_i - \mathbf{g}^*\|, \quad i \in \mathcal{H} \quad (\text{A})$$

$$\|\mathbf{g}_i - \mathcal{A}(\mathcal{G})\| \geq \|\mathbf{g}_i - \mathbf{g}^*\| - \|\mathcal{A}(\mathcal{G}) - \mathbf{g}^*\|, \quad i \notin \mathcal{H} \quad (\text{B})$$

- There are $(1 - \alpha)n$ of (A) and αn of (B). Summing up over i leads to

$$\underbrace{0}_{\text{because of } (\diamond)} \geq \sum_{i=1}^n \|\mathbf{g}_i - \mathcal{A}(\mathcal{G})\| - \sum_{i=1}^n \|\mathbf{g}_i - \mathbf{g}^*\|$$

$$\geq (1 - 2\alpha)n \|\mathcal{A}(\mathcal{G}) - \mathbf{g}^*\| - 2 \sum_{i \in \mathcal{H}} \|\mathbf{g}_i - \mathbf{g}^*\|$$

lead to the theorem since $\frac{1}{(1-\alpha)n} \sum_{i \in \mathcal{H}} \|\mathbf{g}_i - \mathbf{g}^*\| \leq \sigma$



How to achieve smaller aggregation error?

$$\|\mathcal{A}(\mathcal{G}) - \mathbf{g}^*\| \leq C_\alpha \sigma, \quad \text{with } \sigma = \max_{\mathcal{G}} \frac{1}{(1-\alpha)n} \sum_{i \in \mathcal{H}} \|\mathbf{g}_i - \mathbf{g}^*\|.$$

Q: How to achieve a smaller robust aggregation error?

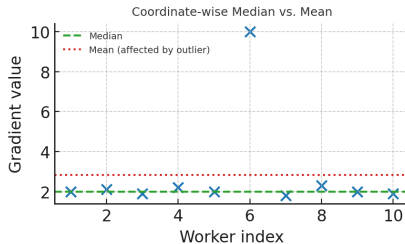
- Fewer Byzantine nodes (smaller α) since $C_\alpha(\text{GeoMed}) = 1 + \frac{1}{1-2\alpha}$
- Reduce the variance of stochastic gradients at each node ([see SVRG, momentum in Lectures 9-10 and the reference below](#))
- Various robust aggregation has the same form of aggregation error bound with different C_α . **Q:** Other aggregations for better C_α ?

Z. Wu, Q. Ling, T. Chen, G. Giannakis "Federated variance-reduced stochastic gradient descent with robustness to byzantine attacks", *TSP 2020*



Example: Coordinate-wise median

$\text{Median}(\{g_i\})$ = element-wise median across all node gradients.



- Resistant to up to 50% Byzantine nodes per coordinate.
- Computationally simple and $C_\alpha = 1 + O(\alpha)$.

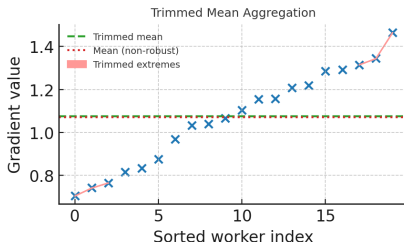
D. Yin, Y. Chen, R. Kannan, P. Bartlett "Byzantine-robust distributed learning: Towards optimal statistical rates", *ICML 2018*



Example: Trimmed mean

For each coordinate, drop top- b largest and top- b smallest nodes, and then average the rest

$$\text{TrimmedMean}(\{g_i\}) = \frac{1}{n - 2b} \sum_{i \in \text{middle } n-2b} g_i$$



Robustness property: If the fraction of Byzantine nodes $\alpha = b/n < \frac{1}{2}$, then the trimmed mean estimator satisfies $C_\alpha = O\left(\frac{1}{(1-2\alpha)^2}\right)$



Lower bound of the robust aggregation error^{*}

Theorem 2 (Fundamental lower bound on robustness)

For any aggregation rule $\mathcal{A}(\mathcal{G})$, there exists $\mathcal{G} = \{\mathbf{g}_1, \dots, \mathbf{g}_n\}$ satisfying

$$\frac{1}{(1-\alpha)n} \sum_{i \in \mathcal{H}} \|\mathbf{g}_i - \mathbf{g}^*\| \leq \sigma, \quad \text{for some } \alpha < \frac{1}{2},$$

such that the robust aggregation output satisfies

$$\|\mathcal{A}(\mathcal{G}) - \mathbf{g}^*\| \geq C_\alpha^* \sigma, \quad \text{with } C_\alpha^* = \frac{\alpha}{1-\alpha}.$$

- **Implication:** Every robust aggregator suffers a linear degradation as the Byzantine fraction α increases. **Q:** How about large σ ?



S. Karimireddy, L. He, M. Jaggi "Byzantine-robust learning on heterogeneous datasets via bucketing", *ICML 2022*

Challenge of robust aggregation under non-i.i.d. data

Design principle so far:

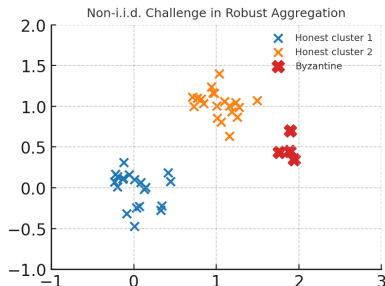
$$\text{Gradient consistency} \quad \mathbf{g}_1 \stackrel{d}{=} \mathbf{g}_2 \stackrel{d}{=} \dots \stackrel{d}{=} \mathbf{g}_n \sim \nabla F(\mathbf{x}; \xi)$$

Challenges:

- Honest nodes may send statistically different gradients.

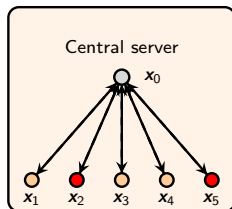
$$\mathbf{g}_1 \stackrel{d}{\neq} \mathbf{g}_2 \stackrel{d}{\neq} \dots \stackrel{d}{\neq} \mathbf{g}_n$$

- Hard to distinguish **malicious gradients** from **natural heterogeneity**.



Robust aggregation based on model consistency

Model consistency $\mathbf{x}_1 = \mathbf{x}_2 = \dots = \mathbf{x}_n \sim \underset{\mathbf{x}}{\operatorname{argmin}} f(\mathbf{x})$



Consensus-based formulation:

$$\min_{\{\mathbf{x}_i \in \mathbb{R}^d\}} \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}_i), \quad \text{s. t. } \mathbf{x}_1 = \mathbf{x}_2 = \dots = \mathbf{x}_n.$$

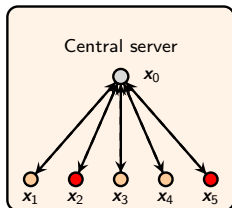
Q: Can we solve this ideal problem if some nodes are Byzantine? **A:** No

$$\min_{\{\mathbf{x}_i \in \mathbb{R}^d\}} \sum_{i \neq \text{Byzantine}} f_i(\mathbf{x}_i), \quad \text{subject to } \mathbf{x}_1 = \dots = \mathbf{x}_i, \text{ for } i \neq \text{Byzantine}.$$



Li, T. Chen, G. Giannakis, Q. Ling, "RSA: Byzantine-robust stochastic aggregation methods for distributed learning from heterogeneous datasets", *AAAI 2019*

Penalty formulation for aggregation under non-i.i.d. data



Penalty-based formulation:

$$\min_{\{\mathbf{x}_i \in \mathbb{R}^d\}} \sum_{i \neq \text{Byzantine}} (f_i(\mathbf{x}_i) + \lambda \|\mathbf{x}_i - \mathbf{x}_0\|_1)$$

when $\lambda \geq \lambda_0$ with some threshold λ_0 , it is equivalent to the constrained formulation.

$$\mathbf{x}_i^{k+1} = \mathbf{x}_i^k - \eta (\nabla F(\mathbf{x}_i^k; \xi_i^k) + \lambda \text{sign}(\mathbf{x}_i^k - \mathbf{x}_0^k)) \quad (\text{Update at Honest})$$

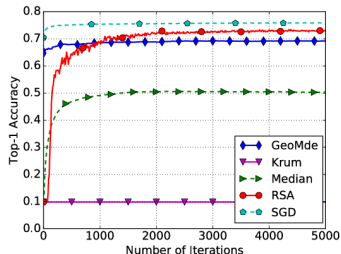
$$\mathbf{x}_0^{k+1} = \mathbf{x}_0^k - \eta \lambda \underbrace{\sum_{i=1}^n \text{sign}(\mathbf{x}_i^k - \mathbf{x}_0^k)}_{\text{subgradient of } \|\mathbf{x}_i - \mathbf{x}_0\|_1} \quad (\text{Global aggregation})$$

include Byzantine

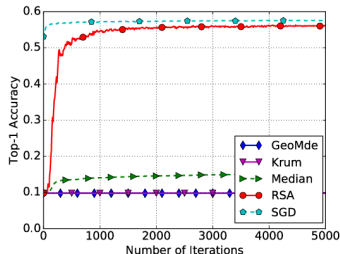


L. Li, T. Chen, G. Giannakis, Q. Ling, "RSA: Byzantine-robust stochastic aggregation methods for distributed learning from heterogeneous datasets", *AAAI 2019*

Performance of robust aggregation under non-i.i.d. data



(a)



(b)

Figure: Top-1 accuracy under attacks on non-i.i.d. data from MNIST dataset with non-uniform distributed labels: (a) $b = 4$; (b) $b = 8$ out of $n = 20$ nodes.



Li, T. Chen, G. Giannakis, Q. Ling, "RSA: Byzantine-robust stochastic aggregation methods for distributed learning from heterogeneous datasets", *AAAI 2019*

Why is the sign-based aggregation Byzantine-robust?

Fact (coordinate-wise majority voting).

Let $\mathcal{B}$ be Byzantine nodes with $|\mathcal{B}| = b$. For coordinate j , define $\Delta_j^k := \mathbf{x}_j^k - \mathbf{x}_0^k$, then for *any* adversarial messages $\{\Delta_{i,j}^k\}_{i \in \mathcal{B}}$, we have

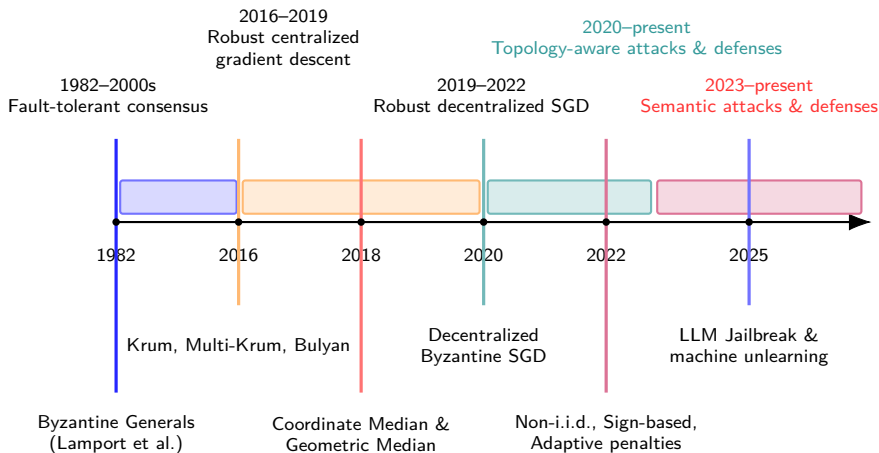
$$\text{sign}\left(\sum_{i=1}^n \text{sign}(\Delta_{i,j}^k)\right) = \text{sign}\left(\sum_{i \in \mathcal{H}} \text{sign}(\Delta_{i,j}^k)\right) \quad \text{if} \quad \boxed{\left|\sum_{i \in \mathcal{H}} \text{sign}(\Delta_{i,j}^k)\right| > b}.$$

Reason: Each Byzantine changes the sum by at most ± 1 per coordinate, so cannot flip the majority if the honest margin exceeds b .

- **Magnitude-free:** Adversaries cannot exploit arbitrarily large updates; only the *sign* counts.



A brief history of Byzantine-robust distributed learning



Evolution of Byzantine-robust distributed learning



Recap of the lecture

- What we have talked about **today**?
 - ⇒ Distributed learning is powerful - but vulnerable to malicious clients.
 - ⇒ Byzantine-robust aggregation improves resilience to attacks.
 - ⇒ Future defense methods should be **semantic** and **provably robust**.



Welcome anonymous survey!

